

UV-Tune: Electrically Tunable AlGa_N-on-Sapphire Metasurfaces for Phase-Encoded Secure Ultraviolet Communication

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1. **Abstract:** Advances in modern communication have escalated demand for secure, high-speed optical links in satellites, although many of these systems struggle under real-world, daytime conditions. Most systems operate at visible or infrared wavelengths where solar background noise and atmospheric turbulence scramble optical phase, overwhelm receivers, and force satellites to reduce data rates or abandon coherent modulation. These restraints create bottlenecks in satellite data transfer that delay remote-sensing operations and secure communications despite increasing launch capacity. Ultraviolet (UV) light offers an intrinsically superior solution via strong atmospheric absorption which creates a solar blind channel with increased security and low noise. However existing UV communication systems rely almost exclusively on intensity modulation, lacking compact, electrically tunable components capable of dynamic phase control. UV-Tune addresses this gap by proposing an electrically tunable AlGa_N-on-sapphire metasurface platform, a wide-bandgap semiconductor alloy optimized for deep-UV operation, that enables phase-encoded ultraviolet communication at the transmitter. Voltage-controlled carrier modulation through Drude dispersion in AlGa_N produces controllable refractive-index shifts resonantly enhanced to stable optical phase states, eliminating reliance on bulky adaptive optics or post-correction schemes. Building upon preliminary simulations, UV-Tune demonstrates how device-level electrical phase control advances system-level satellite performance by reducing bit error rates (BER) under realistic UV noise while enabling higher-speed data transfer. Device-level phase tuning is mapped into a communication framework using finite-difference time-domain simulations and phase-state abstraction. This

approach establishes a modular, open-source pathway toward compact, secure, and turbulence-resilient satellite communication beyond existing UV hardware.

2. The Idea: *As global data traffic and satellite communication grow, atmospheric turbulence distorts optical phase through rapid refractive-index fluctuations and scintillation (Andrews & Phillips, 2005; Kaushal & Kaddoum, 2017). In strong turbulence, BERs can rise from $\sim 3 \times 10^{-4}$ in weak conditions to $\sim 2 \times 10^{-2}$, which is a >6,500% increase, highlighting how refractive-index fluctuations and scintillation severely degrade FSO signal quality and reliability without compensation even at high transmitted power (Xu, Xu, & Zheng, 2021). These impairments worsen at higher data rates during daytime, increasing retransmissions, energy consumption, and system mass—critical constraints for satellites (Chen et al., 2020)—forcing most operational systems to rely on intensity-only modulation. Ultraviolet communication offers advantages like a low solar background, strong atmospheric absorption, and a solar-blind window that enhances security (Xu & Sadler, 2008; Shaw et al., 2013), yet existing UV systems lack compact, electrically tunable devices capable of dynamic phase control, preventing access to the benefits of coherent modulation (Zhang et al., 2019; Zhou et al., 2024). In the visible and infrared, coherence is partially preserved using adaptive optics and digital phase tracking, but these approaches are limited by hardware complexity, power consumption, finite correction bandwidth, and turbulence-induced phase noise (Andrews & Phillips, 2005). *Experimental UV links* using deep-UV LEDs and photomultipliers demonstrate robustness under strong background illumination (Shaw et al., 2013; Xu et al., 2018). They rely almost exclusively on intensity modulation, and other tunable UV approaches are inefficient or poorly integrable at deep-UV wavelengths and leave coherent phase encoding largely inaccessible (Khan et al., 2020; Zhang et al., 2019). Although metasurfaces enable static phase control at longer wavelengths,*

electrically tunable deep-UV metasurfaces remain largely unexplored, leaving UV transmitters fundamentally BER-limited. *To address this*, my work proposes UV-Tune, an electrically tunable AlGaIn-on-sapphire metasurface that enables phase-encoded coherent UV communication by defining stable, electrically controlled phase states directly at the transmitter rather than attempting post-propagation phase correction (Figure 4). Subwavelength AlGaIn resonators support high-Q (damping) modes at deep-UV wavelengths, where applied voltage modulates free-carrier concentration via plasma dispersion, producing small refractive-index shifts that are amplified by resonant field enhancement into large, repeatable optical phase changes without bulky electro-optic components (Ambacher et al., 2002; Rigutti et al., 2019). System-level modeling maps these phase states to a UV communication channel and shows that coherent Binary Phase Shift Keying (BPSK) modulation reduces BER by over two orders of magnitude, with further Q enhancement yielding further improvement (Figures 1a–c), demonstrating that UV-Tune eliminates reliance on adaptive optics and complex digital correction while introducing controllable phase as a new degree of freedom that improves spectral efficiency without sacrificing UV’s intrinsic noise immunity and security (Xu & Sadler, 2008; Chen et al., 2020).

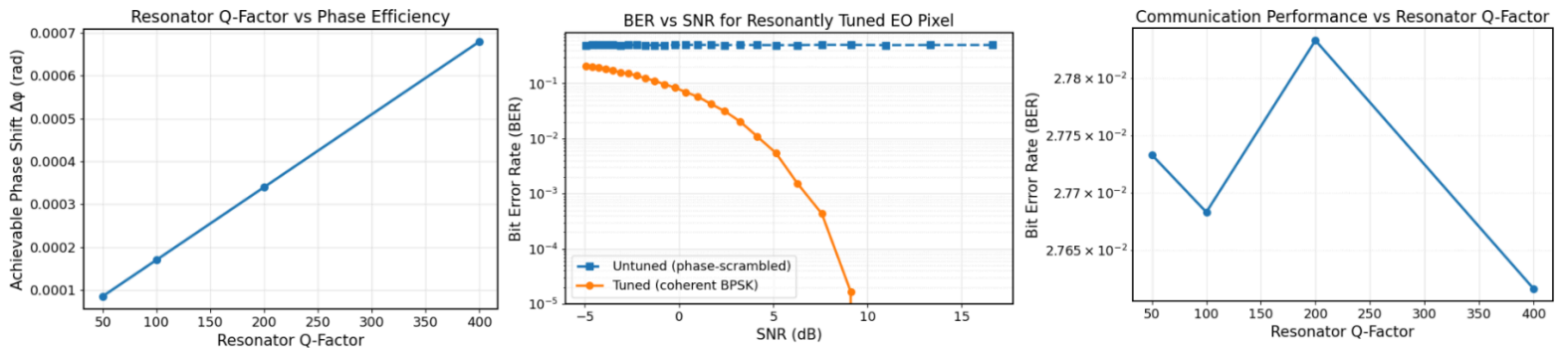


Figure 1. (a) Optical phase shift $\Delta\phi$ increases with resonator Q-factor due to enhanced carrier-induced refractive index

modulation. (b) BER versus SNR shows that electrically tuned metasurfaces significantly outperform untuned devices. (c) BER decreases with increasing Q under fixed noise, indicating improved phase efficiency; minor fluctuations arise from Monte Carlo noise. Together, these results demonstrate that high-Q, electrically tunable AlGaIn metasurfaces enable reliable phase-encoded

ultraviolet communication at low SNR (Data generated by Khanna, 2025).

3. The Plan

3.1. Approach: UV-Tune follows a phased, simulation-driven plan that validates feasibility and de-risks future experimental implementation, incrementally advancing toward a system-relevant UV communication platform. Each phase builds on a AlGa_N metasurface pixel, progressively increasing physical realism and system relevance while maintaining feasibility.

This structure provides a pathway from nanoscale device physics to system-level performance.

Phase 1: Establishing Tunable Metasurface Device Framework and Baseline Device Modeling

This phase establishes the physical basis for electrical phase control in the AlGa_N-on-sapphire metasurface. Prior work shows that changing carrier density through electrooptical gating dynamically controls the optical phase of subwavelength resonators. UV-Tune extends this concept to the deep-ultraviolet using AlGa_N, a wide-bandgap semiconductor compatible with electrical carrier modulation in UV. Simulations model a single AlGa_N metasurface pixel near 270 nm, where an applied voltage alters free-carrier concentration, producing a permittivity change described by a Drude dispersion model (to simulate a Schottky diode). Resonant field enhancement amplifies this effect into a measurable, voltage-dependent optical phase shift.

Under idealized conditions, the model demonstrates binary phase control, confirming the technical feasibility of electrically induced deep-UV phase modulation at the device level.

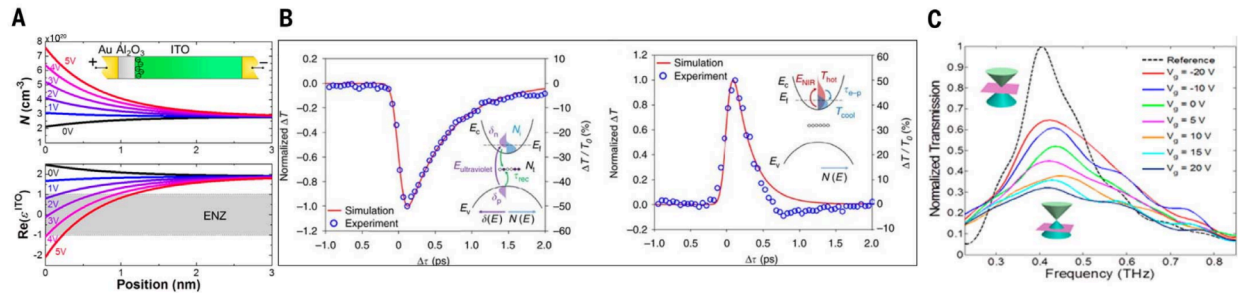
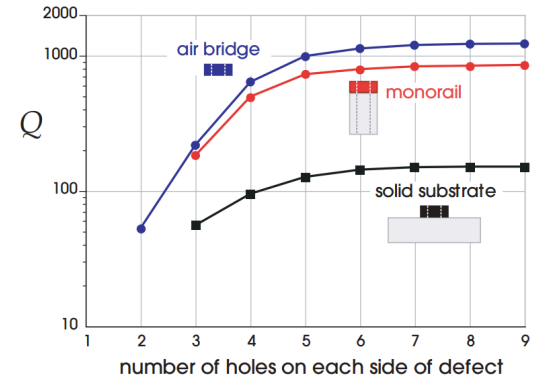


Figure 2. Active tuning in metasurfaces: (A) Voltage-gated ITO metasurface forming nanoscale electron accumulation for phase control; (B) Optically pumped AZO layer altering carrier density and transmission; (C) Graphene-based metasurface where Fermi-level shifts modulate optical response (Shaltout, Shalaev, & Brongersma, 2019).

Phase 2: Introducing Realistic Resonator Losses and Q-Factor Constraints

This phase replaces idealized resonators with loss-aware models including radiation, absorption, and substrate losses. The resonator quality factor (Q) measures how long light remains confined before energy is lost, and determines how sharply the device responds to electrical tuning. Rather than taking a fixed Q , the model sweeps literature-reported values to show resonance broadening and identify thresholds below which electrical tuning fails. This establishes quantitative limits linking resonator Q , voltage-induced permittivity changes, and achievable optical phase modulation, ensuring experimentally realistic figures.

Figure 3. Total Q -factor of a point-defect resonator in a photonic crystal waveguide as a function of the number of holes on either side of the defect. The figure shows three structural variations: air-bridge cavity (top), monorail with $\epsilon = 2.25$ substrate (middle), and solid substrate (bottom). Q saturates for large N to the intrinsic radiation loss, Q_r (Joannopoulos et al., 2011, p. 131).



Phase 3: Transition from Binary to Multi-Level Phase Encoding

With realistic resonator behavior established, the model extends from binary to multi-level electrical phase encoding. Continuous voltage-phase responses are discretized into resolvable phase states, considering carrier saturation and finite linewidth. The approach exploits the device limits defined in Phase 2 with no new physical mechanisms. Usable phase states are set by phase-level separation relative to linewidth and noise, demonstrating that a single electrically tunable UV metasurface pixel can support coherent phase-encoded signaling beyond simple on/off modulation.

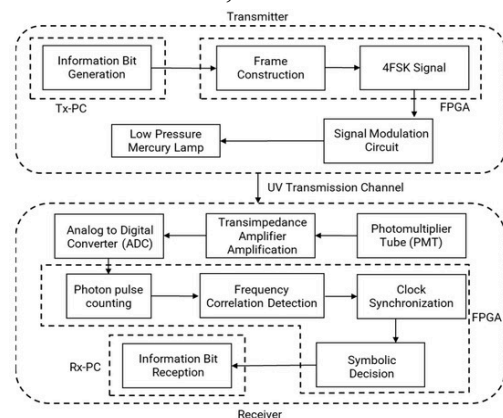
Phase 4: System-Level Integration into an Ultraviolet Communication Channel

The refined metasurface pixel model is embedded in a simplified ultraviolet free-space communication framework, incorporating distance-dependent attenuation and leveraging

deep-UV's solar-blind background. Figure 4 shows signal flow from the phase-modulated transmitter through the propagation channel to the receiver. Performance is quantified via standard signal-to-noise ratio models, with phase-encoded transmission evaluated using phase-sensitive bit-error-rate metrics to compare tuned versus untuned metasurfaces. This establishes a direct quantitative link between nanoscale electrical tuning and end-to-end UV communication. All phases rely on electromagnetic theory, semiconductor electrostatics, and classical communication models.

Figure 4. Block diagram of an ultraviolet weak-signal communication system showing phase-modulated transmission, atmospheric propagation, and receiver detection. (Zhao et al.,

2024)



3.2. Resources: UV-Tune began as an experimental project, but practical challenges necessitated foundational verification first. It is now simulation-driven, using established computational tools and theoretical models to validate electrically tunable AlGaIn metasurfaces before costly fabrication, ensuring physical realism within feasible resources. **Computational and Technical Resources:** Electromagnetic behavior of the metasurface pixel will be modeled using FDTD (MIT MEEP, Lumerical) and frequency-domain solvers (Elmer FEM, FeNiCS). System-level modeling—voltage-dependent phase-state mapping, Monte Carlo noise evaluation, and BER analysis—will be implemented in Python or MATLAB. While system-level simulations are lightweight, device-level FDTD runs will utilize a high-performance workstation. All software is accessible via educational licenses or open-source packages, minimizing logistical risk and adhering to budget constraints.

Mentorship and Expertise: Project guidance will be provided by the MIT THINK Team, offering technical validation and feedback on modeling assumptions while supporting simulation development and data analysis, which will be entirely student led. Mentorship outreach is also

being done to local universities in nanophotonics to help review assumptions and clarify concepts. **Funding:** UV-Tune requires minimal funding, mainly for optional high-performance computing, educational software licenses, and reference materials, all of which can be fully covered by the MIT THINK budget. No specialized laboratory equipment or fabrication is needed, ensuring the project is feasible and financially low-risk.

3.3. Goals: The primary goal of UV-Tune is to demonstrate that electrically tunable AlGaIn-on-sapphire metasurfaces can reliably generate discrete optical phase states for phase-encoded ultraviolet communication, linking device-level electrical control to system-level metrics such as bit error rate (BER). The project follows four incremental milestones with clear completion criteria: **Phase 1, Baseline Device Modeling**, establishes binary phase control under idealized conditions, validated by reproducible voltage-dependent phase shifts matching theoretical expectations. **Phase 2, Realistic Resonator Modeling**, incorporates Q-factor limitations, material losses, and substrate effects, with success defined by physically plausible phase shifts and identification of minimum Q-factor thresholds for effective modulation. **Phase 3, Multi-Level Phase Encoding**, extends the model to 4–8 discrete phase states, ensuring sufficient separation relative to resonator linewidth and channel noise. **Phase 4, System-Level Evaluation**, integrates the metasurface into a simplified UV free-space communication channel, with completion quantified by BER and SNR improvements to confirm coherent phase-encoded signaling. Device-level tests measure phase shifts versus voltage and Q-factor. System-level Monte Carlo BER simulations assess performance under noise. Success requires reproducible phase shifts, resolvable states, and statistically significant BER improvement. Targets include $\geq \pi$ radians phase shift per resonator, a minimum of four resolvable phase states scalable with Q-factor, and at least an order of magnitude BER improvement relative to untuned transmission.

All simulations, modeling, and analysis will be conducted independently, with guidance from the MIT THINK Team and periodic external mentor input to ensure technical accuracy.

3.4. Risks: *1. High computational demand:* Simulating high-Q AlGaIn metasurfaces at deep-UV wavelengths is time-intensive. To mitigate this, the project will scale up gradually from single-unit-cell models and employ parallelized open-source FDTD solvers such as MEEP. To ensure accuracy without excessive computation we will use frequency-domain solvers like MPB or COMSOL. *2. Uncertainty in deep-UV material parameters:* Variations in AlGaIn refractive index or carrier mobility could affect predicted phase shifts and BER. The project will use literature-based parameter sweeps and perform sensitivity analyses to quantify their impact, ensuring robust system-level predictions. *3. Multi-level phase encoding limitations:* Resonator linewidth, voltage resolution, or channel noise may limit distinguishable phase states. Mitigation includes validating binary control first, incrementally increasing phase levels, optimizing resonator Q-factors, and using Monte Carlo simulations to evaluate noise robustness

3.5. Timeline: The UV-Tune project will be executed over **8–10 weeks**, with each phase corresponding to a key milestone. **Phase 1 (Weeks 1–2)** establishes a baseline device modeling framework through FDTD simulations of a single AlGaIn metasurface, documenting achievable phase shifts and thresholds for effective electrical tuning in updated simulation files. **Phase 2 (Weeks 3–4)** introduces realistic resonator modeling, including Q-factor limitations, material losses, and substrate effects, with achievable phase shifts and tuning thresholds documented in plots. **Phase 3 (Weeks 5–6)** extends to multi-level phase encoding, creating 4-8 discrete phase states while accounting for linewidth, carrier saturation, and noise. Progress will be tracked with phase-state charts, Monte Carlo simulation data, and version controlled code. **Phase 4 (Weeks 7–8)** integrates the metasurface pixel into a simplified UV communication

system, evaluating system metrics like BER and signal-to-noise ratio, and documenting results with plots and channel assumptions. **Weeks 9–10** are reserved for final analysis, synthesis, and preparation of the report and figures. *Throughout all phases, code, data, and logs will be maintained in a structured repository to ensure clear documentation and reproducibility.*

3.6. Current Progress and Need for Funding: Previously, the UV-Tune project has developed a robust simulation framework linking device-level physics to system-level UV communication performance. My elementary simulation (Github: [UVTune](#)) models a AlGa_N-on-sapphire metasurface pixel at 270 nm with Meep FDTD and realistic geometry, PML boundaries, and material properties, mapping voltage to carrier density up to 5 Volts and Drude-based permittivity shifts. The simulation shows that higher-Q resonators produce larger phase shifts, and system-level Monte Carlo BER analysis under coherent BPSK demonstrates that electrically tuned metasurfaces outperform untuned pixels, directly linking resonator Q-factor to communication performance to validate the project's approach (Figures 1a-c). Preliminary results demonstrate feasibility but remain intentionally non-final, making this project well-suited for iterative THINK mentorship. Refinements—including full complex transmission extraction, improved FDTD resolution, substrate thickness, accurate Drude permittivity, and realistic carrier distributions—are needed for fully quantitative predictions and to capture subwavelength and finite-array effects. BER modeling requires amplitude variations, realistic SNR, and phase noise. Mentorship and funding from the MIT THINK Scholars Program will enable the project to utilize more advanced algorithms and perform extensive parameter sweeps for converged 3D FDTD simulations of larger arrays. Optional Lumerical or COMSOL licenses would allow UV-Tune to move from qualitative trends to physically accurate, system-relevant predictions, linking electrically tunable devices to rapid, phase-encoded UV communication.

3.7. Project Budget (Projected Total Assuming Open Source Software: \$881.98)

Item, Quantity, and Description	Total Cost	Supplier / Link (Example)
High-Performance PC / Workstation (x1) <i>Desktop with ≥ 8-core CPU for fast FDTD simulations</i>	\$849.99 (A high-performance workstation is required to run memory-intensive 3D FDTD simulations locally within the project's time and budget constraints.)	Processor Link Walmart, VprTech Stryker 3.0 Gaming PC - AMD Ryzen 7 Graphic Card:NVIDIA RTX 3060 12GB
FDTD Software (MEEP or Lumerical) (x1) <i>Software for EM simulations</i>	MEEP with MPB \$0 (Open-Source) Lumerical (Educational)	MEEP MPB Link Ansys Lumerical Link
Finite Element Method Software (x1) <i>FDS for phase mapping [Optional]</i>	Elmer FEM / FeniCS \$0 (Open-Source) COSMOL (Educational)	Elmer FEM Link FeniCS Link COSMOL Link
Data Storage / Backup (x1) <i>External SSD or cloud storage</i>	\$31.99	SSD Link Verbatim 256GB Store 'N' Go USB 3.2 Gen 1 Portable SSD

4. Personal Interest: My academic background is rooted in physics, with a focus on optics and electromagnetism. I first discovered optics three years ago through Science Olympiad, competing and medaling at the state and national levels while strengthening my fundamental knowledge. Over the past year, I conducted independent work in photonics, developing simulations in MATLAB for Fresnel refraction, Stokes and Jones polarization, Raman scattering, and negative-index metamaterials (GitHub: <https://github.com/MaanuG>). This work sparked my interest in computational electromagnetics as a bridge between fundamental physics and real systems. My interest in ultraviolet nanophotonics emerged while studying circuits, leading me to explore electrically tunable semiconductor devices and diodes for secure optical communication and to develop the initial concept for UV-Tune. I have strong fundamentals in electromagnetics, photonics, and computational modeling, but require mentorship from the MIT THINK Scholars Program to develop expertise in FDTD and FEM simulations, realistic material modeling, and device-to-system performance analysis to refine my models to a research-quality level.

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